

LECTURE 7: BEST RESPONSE AND MIXED STRATEGIES

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WEEK 3

LET'S THINK ABOUT OUR DECISIONS IN TERMS OF THE DECISIONS OF THE OTHER PLAYS

	Anna buys a small car	Anna buys a big car
Beth buys a small car	Anna 10% probability of dying Beth 10% probability of dying	A: 5% B: 30%
Beth buys a big car car	A: 30% B: 5%	A: 15% B: 15%

We want to construct a **best response** function for Anna and Beth:

For Anna this tells us, what should Anna do based on what Beth does; and for Beth, what should Beth do based on what Anna does.

LET'S THINK ABOUT OUR DECISIONS IN TERMS OF THE DECISIONS OF THE OTHER PLAYS

Let's call the decisions of Anna and Beth, D_A and D_B

Anna's best response function given Beth's decision is:

$$BR_A(D_B) = \begin{cases} \text{Buy a big car} & D_B = \text{Buy a small car} \\ \text{Buy a big car} & D_B = \text{Buy a big car} \end{cases}$$

Here, we say the Anna buying a big car is **dominating**, Anna would choose it no matter what Beth does

Incredibly important: Our best response is shown in terms of the other person's decisions - this is the key part of game theory/strategic thinking. We always want to phrase our actions conditional on the other players' actions

WHEN BEST RESPONSE FUNCTIONS COINCIDE, WE GET A NASH EQUILIBRIUM

$$BR_A(D_B) = \begin{cases} \text{Buy a big car} & D_B = \text{Buy a small car} \\ \text{Buy a big car} & D_B = \text{Buy a big car} \end{cases}$$

$$BR_B(D_A) = \begin{cases} \text{Buy a big car} & D_A = \text{Buy a small car} \\ \text{Buy a big car} & D_A = \text{Buy a big car} \end{cases}$$

Start with one of Anna's decisions → Find Beth's best response → Find Anna's best response (does Anna want to deviate from the original decision):

Example 1:

✓ Anna buys a big car → Beth's best response to buy a big car → Anna's best response to buy a big car

Anna does not deviate from the original decision, so both buying a big car is a Nash equilibrium

WHEN BEST RESPONSE FUNCTIONS COINCIDE, WE GET A NASH EQUILIBRIUM

$$BR_A(D_B) = \begin{cases} \text{Buy a big car} & D_B = \text{Buy a small car} \\ \text{Buy a big car} & D_B = \text{Buy a big car} \end{cases}$$

$$BR_B(D_A) = \begin{cases} \text{Buy a big car} & D_A = \text{Buy a small car} \\ \text{Buy a big car} & D_A = \text{Buy a big car} \end{cases}$$

Start with one of Anna's decisions → Find Beth's best response → Find Anna's best response (does Anna want to deviate from the original decision):

Example 2:

✗ Anna buys a small car → Beth's best response to buy a big car → Anna's best response to buy a big car

Anna **does** deviate from the original decision, so Anna buying a small car is not part of a Nash equilibrium

DOMINATING STRATEGIES

If a player would always choose one particular strategy, we call that strategy dominating

- If, depending on the other players action, we'd be indifferent between the dominating strategy and another, we call it **weakly dominating**

- If the dominating strategy is always strictly better than any other action, we call it **strictly dominating**

- If one strategy is strictly dominating, the others are **dominated**. Players should never play these
- Finding dominated strategies can help us solve games quicker by eliminating options

FINDING NASH EQUILIBRIA WITH DOMINATED STRATEGIES

Remember our car safety problem. Buying a small car is **dominated** for both people.

	Anna buys a small car	Anna buys a big car
Beth buys a small car	Anna 10% probability of dying Beth 10% probability of dying	A: 5% B: 30%
Beth buys a big car car	A: 30% B: 5%	A: 15% B: 15%

SOMETIMES THERE ARE ONLY NASH EQUILIBRIA IF PLAYERS RANDOMIZE

	P1 scissors	P1 paper	P1 rock
P2 scissors	Draw	P2 wins	P1 wins
P2 paper	P1 wins	Draw	P2 wins
P2 rock	P2 wins	P1 wins	Draw

Let's say that people win \$1 from winning, and \$0 otherwise

WE CAN USE OUR BEST RESPONSE FUNCTION TO CONVINCE OURSELVES THERE ARE NO PURE NASH EQUILIBRIA

$$BR_i(D_j) = \begin{cases} \text{Scissors} & D_j = \text{Paper} \\ \text{Paper} & D_j = \text{Rock} \\ \text{Rock} & D_j = \text{Scissors} \end{cases}$$

- ✗ P1 plays rock → P2's best response to play paper → P1 would want to deviate to scissors
- ✗ P1 plays scissors → P2's best response to play rock → P1 would want to deviate to paper
- ✗ P1 plays paper → P2's best response to play scissors → P1 would want to deviate to rock

If players respond to each other, then they will always want to deviate

BUT IF PLAYERS RANDOMIZE, THEN THERE IS A NASH EQUILIBRIUM

We call a Nash equilibrium mixed, if players make decisions with some probability

If each player plays each move with $\frac{1}{3}$ probability, there is a mixed strategy equilibrium

	P1 scissors	P1 paper	P1 rock
P2 scissors	Draw with 1/9 prob	P2 wins with 1/9 prob	P1 wins with 1/9 prob
P2 paper	P1 wins with 1/9 prob	Draw with 1/9 prob	P2 wins with 1/9 prob
P2 rock	P2 wins with 1/9 prob	P1 wins with 1/9 prob	Draw with 1/9 prob

Each payoff has a $\frac{1}{9}$ chance of happening. The expected value is the sum over the entry game matrix - P1 wins $\frac{3}{9}$ of the time; P2 wins $\frac{3}{9}$ of the time; draw $\frac{3}{9}$ of the time

LET'S LEARN TO CALCULATE THESE PROBABILITIES IN A SIMPLER GAME

There are two firms looking to enter a market. If both enter, they both make \$-1 profit, if only one enters then that firm makes \$1 profit, otherwise they both make no money

	Firm 1 enters	Firm 1 does not enter
Firm 2 enters	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter	F1 profit = 1 F2 profit = 0	F1 profit = 0 F2 profit = 0

Let p_1 be the probability that firm 1 enters, and p_2 the probability that firm 2 enters

Our goal: Find these probabilities!

LET'S LEARN TO CALCULATE THESE PROBABILITIES IN A SIMPLER GAME

Our Nash equilibrium condition with randomization is that we wouldn't deviate given the probability expectations of the other person's actions

We're indifferent between our actions on average

The expected payoff for Firm 1 entering should equal the expected payoff for Firm 1 not entering

$$E(\pi_1 | \text{Firm 1 enters}) = E(\pi_1 | \text{Firm 1 does not enter})$$

That is, the expected profit of Firm 1 entering is equal conditional on it entering or not

Tricky part: We use Firm 1's expected profit to find **Firm 2's probabilities**

$$\underbrace{E(\pi_1 | \text{Firm 1 enters})}_{\text{F1 expected payoff if F1 enters}} = \underbrace{p_2 \times -1}_{\text{Expected profit if F2 enters}} + \underbrace{(1 - p_2) \times 1}_{\text{Expected profit if F2 does not enter}}$$

And if Firm 1 does not enter:

$$E(\pi_1 | \text{Firm 1 does not enter}) = 0$$

LET'S LEARN TO CALCULATE THESE PROBABILITIES IN A SIMPLER GAME

$$\begin{aligned} E(\pi_1 | \text{Firm 1 enters}) &= E(\pi_1 | \text{Firm 1 does not enter}) \\ \underbrace{p_2 \times -1}_{\text{Expected profit if F2 enters}} + \underbrace{(1 - p_2) \times 1}_{\text{Expected profit if F2 does not enter}} &= \underbrace{0}_{\text{F1 profit is zero if it does not enter}} \\ -p_2 + 1 - p_2 &= 0 \\ p_2 &= \frac{1}{2} \end{aligned}$$

So the probability of Firm 2 entering is 50%

Again, really tricky we use the expectations of Firm 1's profits to find Firm 2's probabilities

Why? We need Firm 1 to be indifferent between what it chooses - otherwise it would have an incentive to deviate and we wouldn't have a Nash equilibrium

HOW TO THINK ABOUT THESE PROBABILITIES

Let's go back to our scissors paper rock example:

- There's obviously no move that's strictly speaking better than the other
- The best way to play is through randomising your moves - choosing each move with $\frac{1}{3}$ probabilities
- **Why is this a Nash Equilibrium?** If you were to play rock too often - eg 50% of the time - your opponent could statistically take advantage of this anomaly and beat you on average
- So you would want to deviate back to equal probabilities

LET'S TRY ANOTHER EXAMPLE WITH DOMINATING PROBABILITIES

In this case, Firm 1 has entering as a strictly dominating strategy - so it should always play this, even in mixed strategy.

	Firm 1 enters	Firm 1 does not enter
Firm 2 enters	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter	F1 profit = 2 F2 profit = 2	F1 profit = 0 F2 profit = 0

IN MANY REAL GAMES, EQUILIBRIA ONLY EXIST IN MIXED STRATEGIES

In baseball, the pitcher needs to choose whether to pitch high or low; and often the batter needs to commit before the ball is thrown.

Say the pitcher values a strike out as much as a batter values a run. The batter will score 2 runs if the pitch is low and they hit it, and 1 run if its high and they hit it

	Pitcher goes high	Pitcher goes low
Batter goes high	Pitcher: 0 Batter: 1	Pitcher: 1 Batter: 0
Batter goes low	Pitcher: 1 Batter: 0	Pitcher: 0 Batter: 2

1. We'll show there are no pure strategy equilibria
2. We'll find the mixed strategies

IN MANY REAL GAMES, EQUILIBRIA ONLY EXIST IN MIXED STRATEGIES

Our mixed strategy equilibria is:

$$\text{Pitcher} = \left(\begin{array}{cc} \text{Low} & \text{High} \\ \frac{1}{3} & , \quad \frac{2}{3} \end{array} \right)$$

$$\text{Batter} = \left(\begin{array}{cc} \text{Low} & \text{High} \\ \frac{1}{2} & , \quad \frac{1}{2} \end{array} \right)$$

- The pitcher knows that the batter values a high hit more, so it throws fewer high balls
- The batter knows high balls will be less frequent, but are more valuable, the optimal point is to swing high and low with equal probability

